CLIMATE CHANGE

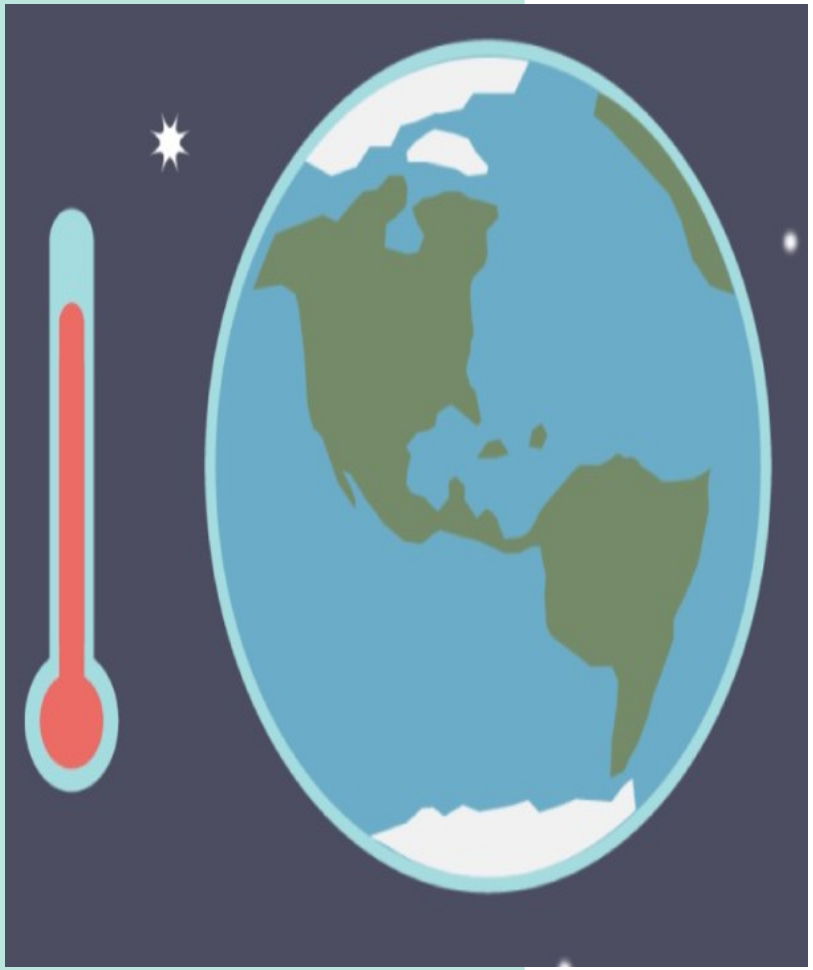


WILL IT BE OUR END?

TOPICS

- WHAT IT **MEANS**
- **GLOBAL WARMING** AND ITS **CAUSES**
- CLIMATE CHANGE & ITS CAUSES
- **IMPACT** ON THE OVERALL ENVIRONMENT
- **MEASURES** TO BE TAKEN





CLIMATE

Weather: The overall condition of a place over a short period of time, considering factors like temperature, rainfall, etc.

Climate: Average of the weather of a particular region

- Measured over a long time, for example, 30 years
- A huge area is taken into consideration; it can be a country, a set of countries, or an entire continent
- Doesn't change frequently







CLIMATE CHANGE

Significant changes in the climate that –

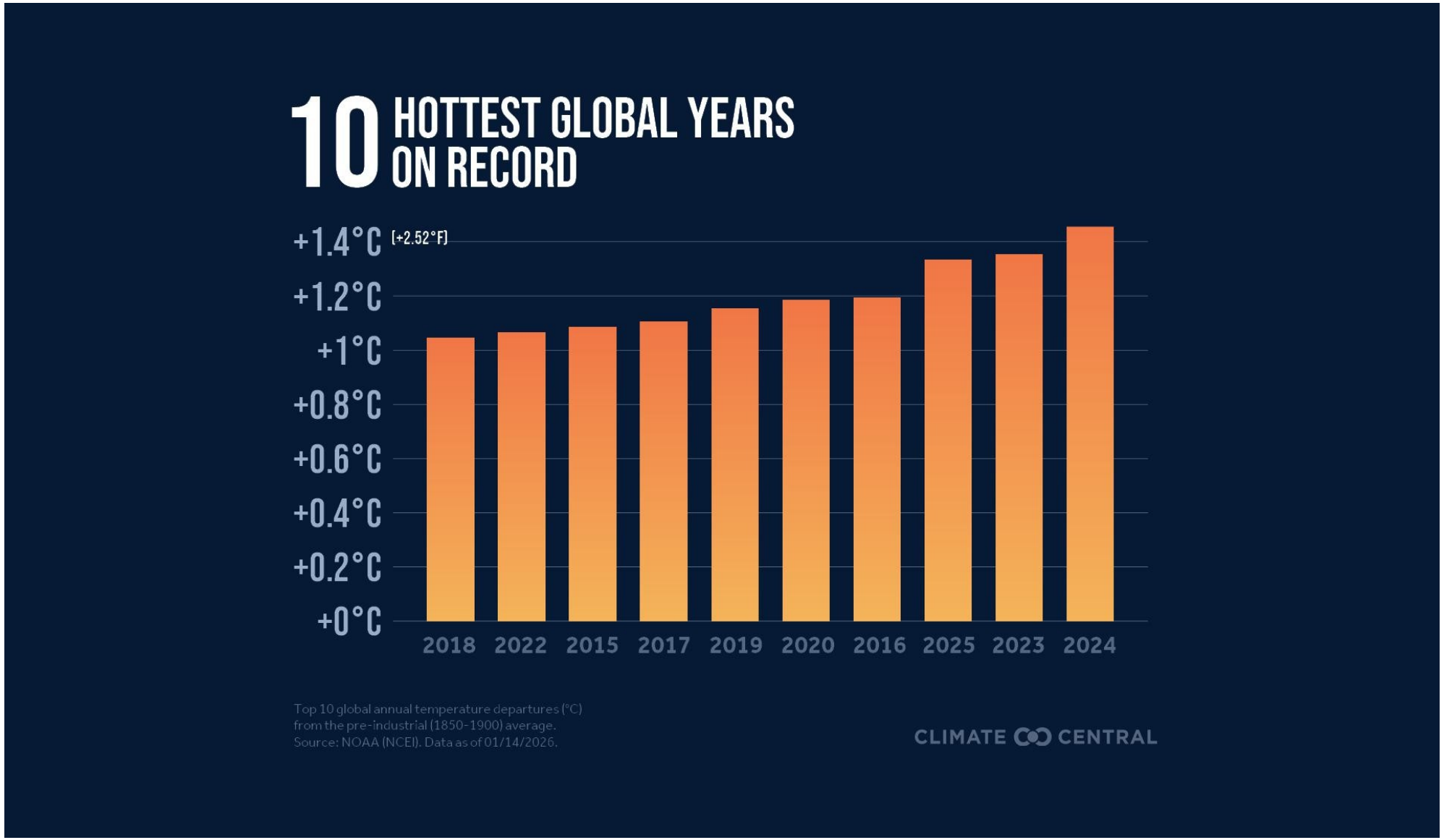
- Affect the overall conditions of a region, such as rainfall, temperature, and humidity.
- Lasts for a long time
- Hampers the natural balance of the environment
- Slowly destroys the biodiversity



WHAT IS GLOBAL WARMING?

Global Warming is the **long-term heating of the Earth's surface** due to various human activities.

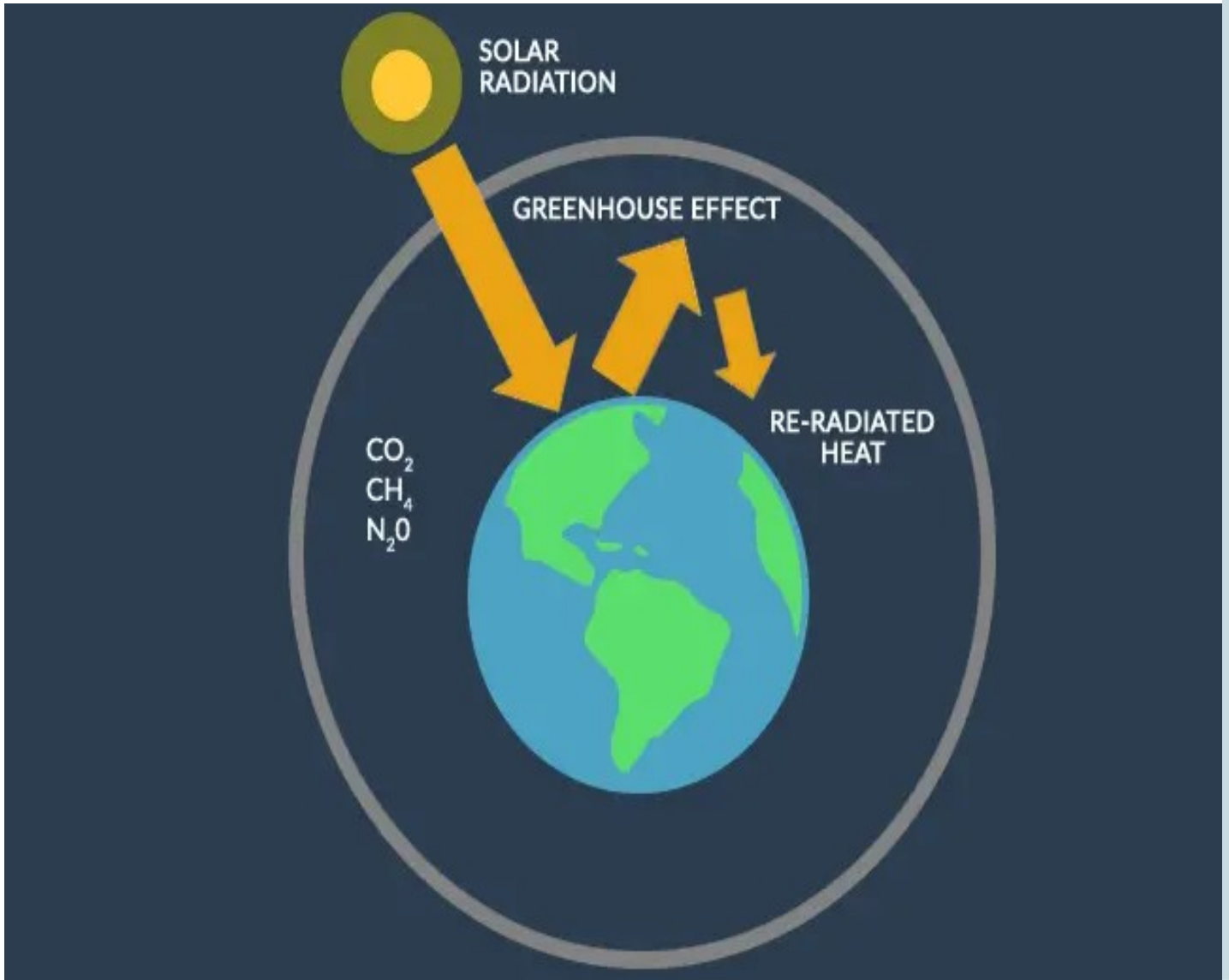
- Sometimes Global Warming and Climate Change are **used interchangeably but they are different**
- Due to climate change, various changes are occurring and one of them is that the Earth's temperature is increasing
- To put it into simple terms, Climate Change is the **cause,** and Global Warming is the **result**

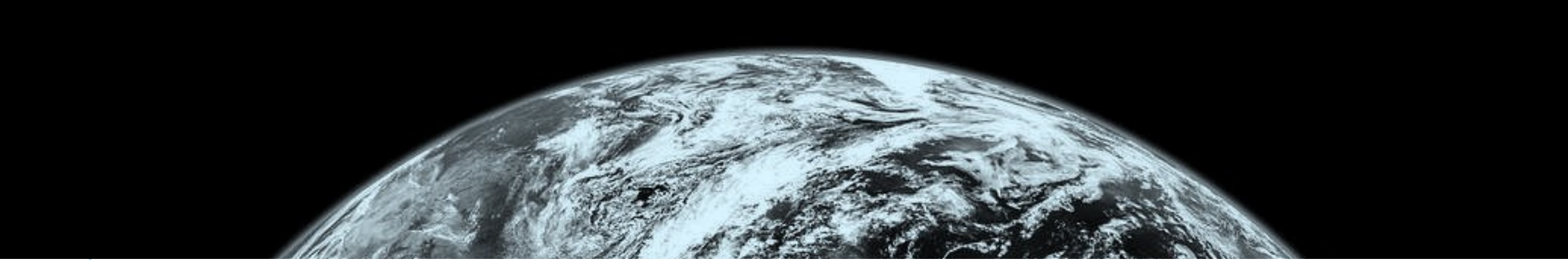

10 HOTTEST GLOBAL YEARS ON RECORD
+1.4°C (+2.52°F)
+1.2°C
+1°C
+0.8°C
+0.6°C
+0.4°C
+0.2°C
+0°C
2018
2022
2015
2017
2019
2020
2016
2025
2023
2024
Top 10 global annual temperature departures (°C) from the pre-industrial (1850-1900) average.
Source: NOAA (NCEI). Data as of 01/14/2026.
CLIMATE CENTRAL

GREENHOUSE EFFECT AND CLIMATE CHANGE

THE GREENHOUSE EFFECT IS THE WAY IN WHICH HEAT IS TRAPPED CLOSE TO THE EARTH'S SURFACE BY GREENHOUSE GASES.

GHGS INCLUDE CO2, CH4, NO2, WATER VAPOR ETC.





STEP 1

During the day, solar radiation reaches the Earth's atmosphere and a portion of it is reflected to space

STEP 2

Some of the heat is absorbed by the land and oceans

STEP 3

At night, starts radiating heat that goes towards the upper atmosphere but is trapped by gases

STEP 4

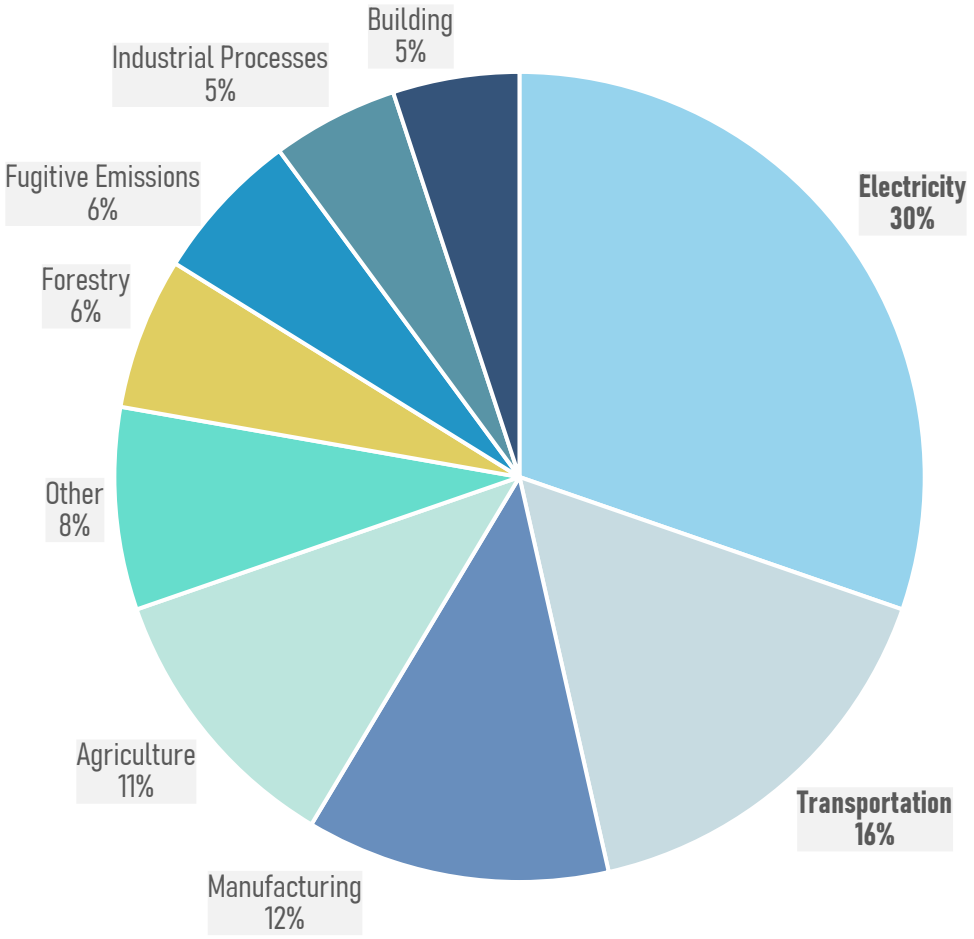
This excessive heat trapping causes the Earth's temperature to rise over time

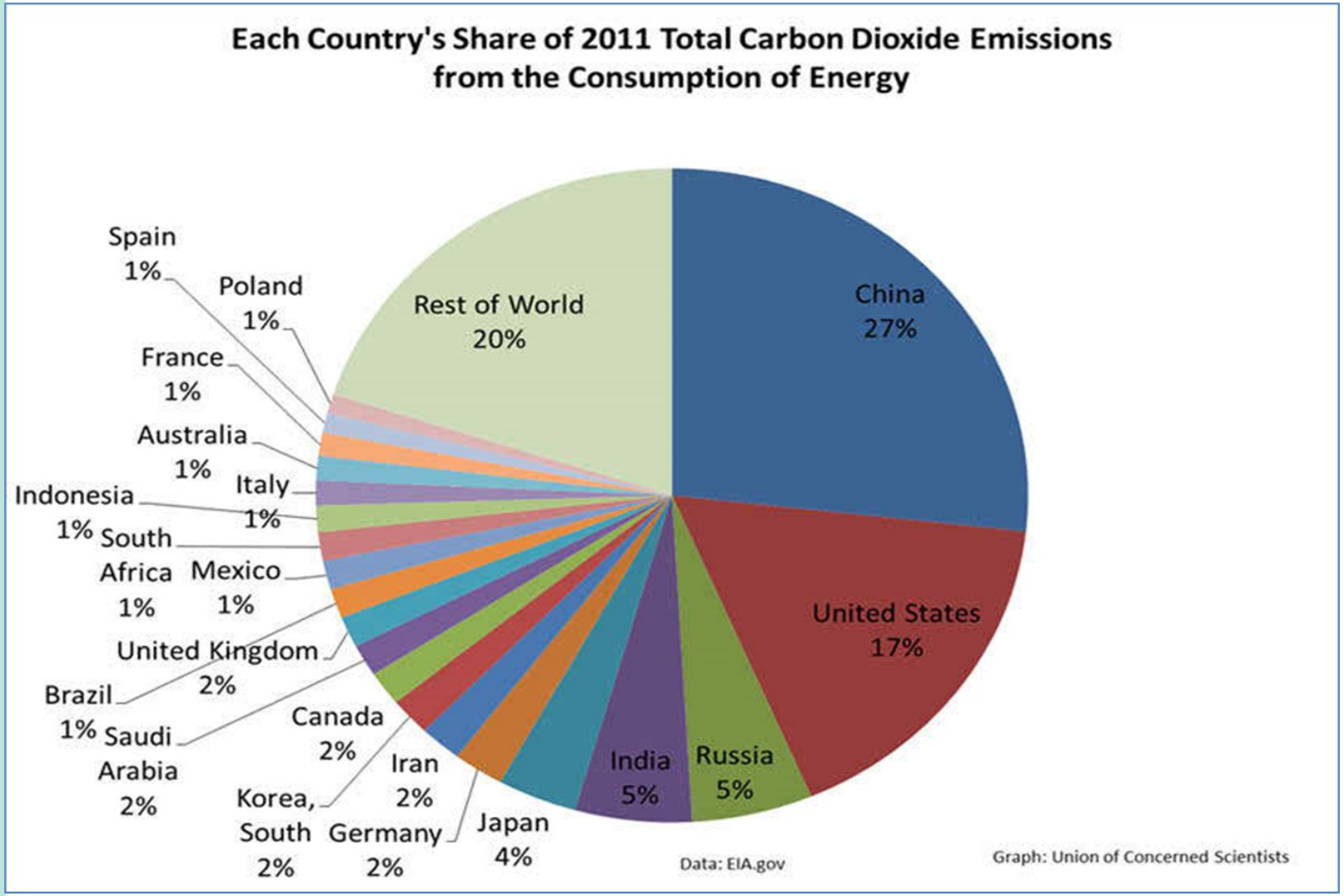
STEP 5

Human activities like burning fossil fuels increase GHGs and the greenhouse effect continues ever so rapidly

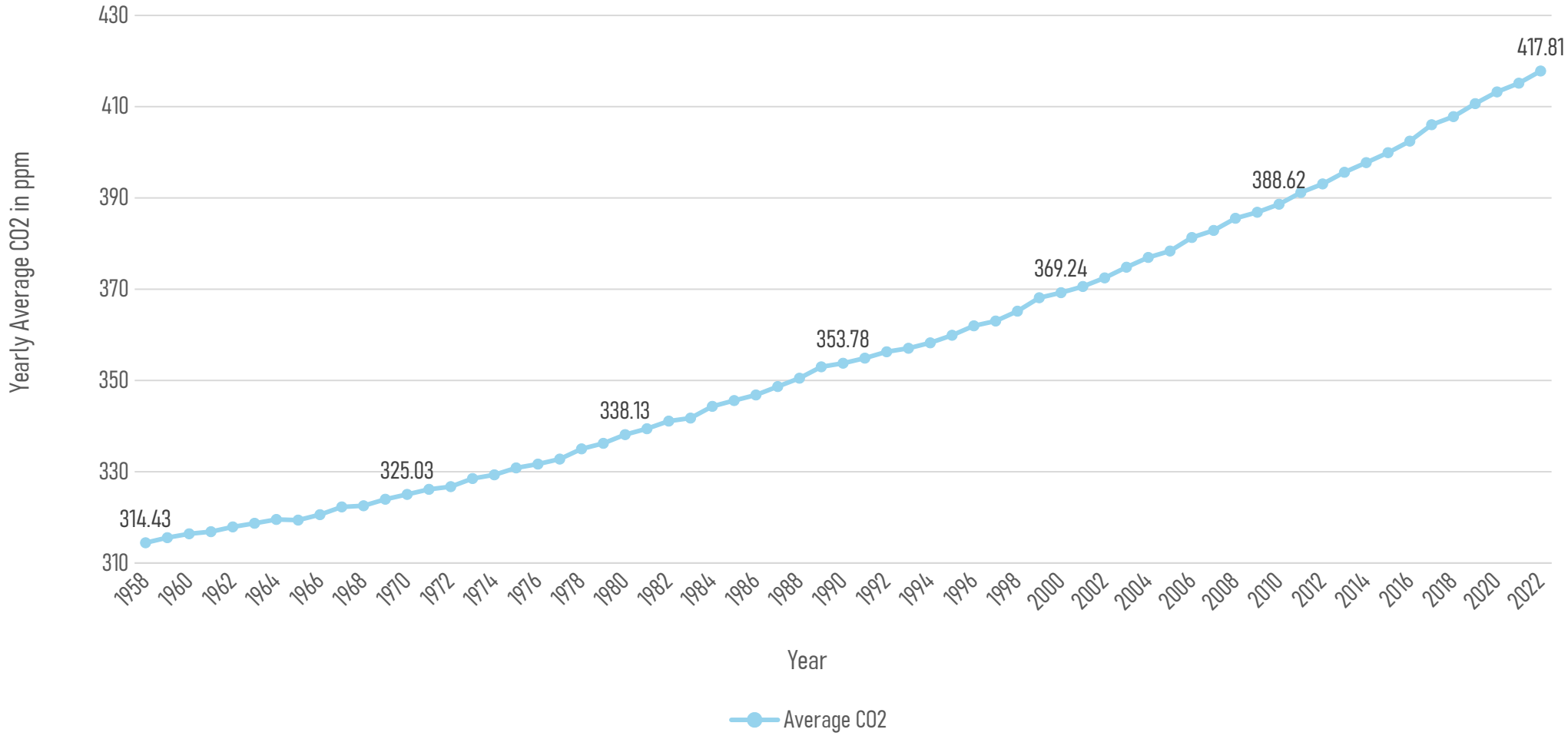
HOW IT HAPPENS

GHG EMISSIONS BY SECTOR




Each Country's Share of 2011 Total Carbon Dioxide Emissions from the Consumption of Energy
China 27%
United States 17%
Russia 5%
India 5%
Japan 4%
Germany 2%
Korea, South 2%
Iran 2%
Canada 2%
Saudi Arabia 2%
United Kingdom 2%
Brazil 1%
Mexico 1%
South Africa 1%
Indonesia 1%
Italy 1%
Australia 1%
France 1%
Poland 1%
Spain 1%
Rest of World 20%
Data: EIA.gov
Graph: Union of Concerned Scientists

INCREASE OF C02 BY YEARS



INTERNATIONAL PROTOCOLS ADOPTED FOR GREEN HOUSE GAS EMISSION

The **UNFCCC** (United Nations Framework Convention on Climate Change) is an international environmental **treaty** that states the **goal of the Kyoto Protocol** as the "stabilization of greenhouse gas concentrations in the atmosphere at a level that would prevent dangerous anthropogenic interference with the climate system."

Why did they make the Kyoto Protocol?

Under the Kyoto Protocol, **industrialized nations agreed to cut their greenhouse gas emissions to a certain percentage below 1990 levels.** The year 1990 was chosen as a baseline because that was the year the UN first launched negotiations on climate change.

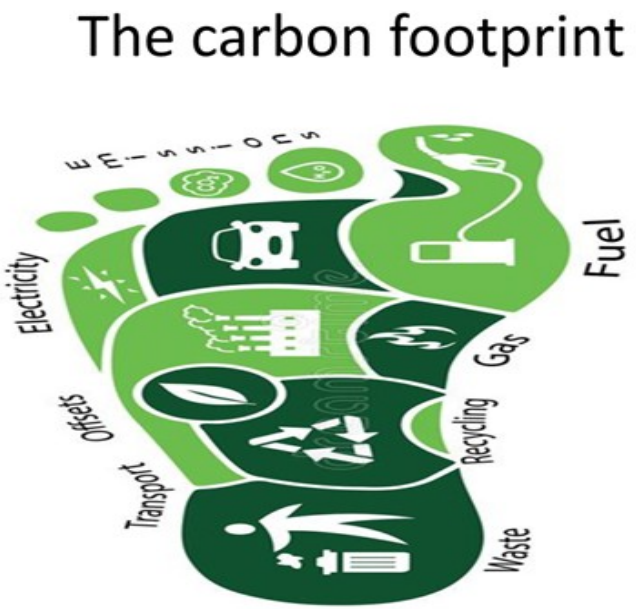
Kyoto Protocol

The Protocol was adopted by Parties to the UNFCCC in 1997 and entered into force in 2005.

The Protocol "recognizes that developed countries are principally responsible for the current high levels of GHG emissions in the atmosphere because of more than 150 years of industrial activity and places a heavier burden on developed nations under the principle of 'common but differentiated responsibilities "

Carbon Footprint

- The term "**carbon footprint**" is often used as shorthand for **carbon (usually in tones) being emitted from organization.**



CARBON TRADING

Carbon Trading is a scheme where firms (or countries) buy and sell carbon permits as part of a program to reduce carbon emissions. Usually, firms are given a certain quota to pollute a certain amount. If they wish to pollute more than their allowance, then they must buy more permits. If they pollute less than their quota, they can sell their spare permits on the market. Thus, there is an incentive to reduce pollution and find the most efficient way of dealing with pollution.

EXAMPLES OF CARBON TRADING

The European Union Emission Trading Scheme (or EU ETS) is the largest multinational greenhouse gas emissions trading scheme in the world. It is one of the EU's central policy instruments to meet its capacity set in the Kyoto Protocol (Jones et al., 2007, p. 64)

Additionally, carbon Trading depends on :

Carbon market

Business-to-business

Regional trade

Country to country

ECOLOGICAL FOOTPRINT

An Ecological Footprint is a measure of the amount of bio-productive land and sea required to support a person's lifestyle. It includes the land needed to grow their food, dispose of their waste, and absorb their carbon emissions. The footprint counts all the impacts of personal spending as well as the business and government expenditure on their behalf. The Ecological Footprint shows how carbon emissions compare and interact with other elements of human demand, such as our pressure on food sources, the quantity of living resources required to make the goods we consume, and the amount of land we take out of production when we pave it over to build cities and roads.

Climate change, deforestation, overgrazing, fisheries collapse, food insecurity, and rapid extinction of species are all part of a single, overarching problem: Humanity is simply demanding more from the Earth than it can provide

THE SEVE (7) THEORIES OF CLIMATE CHANGE

- IT IS WELL-ESTABLISHED THAT CLIMATE CHANGE IS HAPPENING

SCIENTISTS HAVE PRODUCED 7 THEORIES OF CLIMATE CHANGE THAT COMBINE **NATURAL AND HUMAN-DEPENDENT CAUSES**

LET'S SEE WHAT THEY ARE...



THE SEVEN (7) THEORIES

1. **Anthropogenic Global Warming(AGW) theory:** This theory of climate change contends that human emissions of greenhouse gases, principally carbon dioxide (CO2, methane, and nitrous oxide, are causing a catastrophic rise in global temperatures. The mechanism whereby this happens is called **the enhanced greenhouse effect**. It includes the activities of human beings. For example,
    - Usage of Non-renewable energy resources
    - Usage of artificial fertilizers and insecticides
    - General environmental pollution
2. **Human Forcing Besides Greenhouse Gases:** Human activities that directly affect the Earth's surface, like building cities and deforestation, coastal developments, jet contrails, etc., for transformation of the Earth's surface.

- **3. Bio-thermostat theory:** Rising temperatures and CO2 levels trigger biological and chemical responses in the atmosphere that have a cooling effect. The more CO2 there is in the air, the better plants grow, and the more CO2 they remove from the air and store in their leaves, branches, trunks, and roots, as well as in the soil beneath the plants – a suite of processes called "sequestration." Higher temperatures also increase carbon sequestration rates: Carbonyl sulphide (COS), diffuse light, iodo compounds, dimethyl sulphide, and other aerosols.
- **4. Cloud Formation & Albedo:** postulates that changes in the formation and albedo of clouds create negative feedbacks that cancel out all or nearly all the warming effect of higher levels of CO2.How the solar reflectivity of snow has an impact on climate change. **Albedo** is a fraction of light reflected by a body or surface. It is commonly used in astronomy to describe the reflective properties of planets, satellites, and asteroids. Albedo = reflected light/Incoming light %. Earth's planetary albedo is about 31%. Albedo affects global climate and weather patterns. In 1999, Yogesh Sud, a NASA scientist, and his colleagues found that changes in cloud coverage in the tropics acted as a natural thermostat to keep sea surface temperature (SST) between approximately 28°C and 30°C.

THE SEVEN (7) THEORIES

5. **Ocean Currents:** due to the slowdown of the ocean's Thermohaline Circulation (THC). Explain global warming through the energy concentration at the upper layer of the oceans, dependent on temperature variations, which give rise to changes in salt content and hence in density.

5. **Planetary motion:** How the Earth's tilt, orbit, and other planets affect the climate, by natural gravitational and magnetic oscillations of the solar system induced by the planet's movement through space. These oscillations modulate solar variations and/or other extraterrestrial influences on Earth, which then drive climate change

5. **Solar Variability:** How changes in the coronal ejections and magnetic fields of the Sun cause changes in cloud formation, ocean currents, wind, etc., and **cause climate change**. Changes in the brightness of the sun are caused by sunspots – bursts of energetic particles and radiation – that vary in frequency in cycles of roughly 11, 87, and 210 years. These cycles cause changes in the amount of electromagnetic radiation – also called "solar wind" – that reaches Earth and its atmosphere, which in turn affects Earth's climate

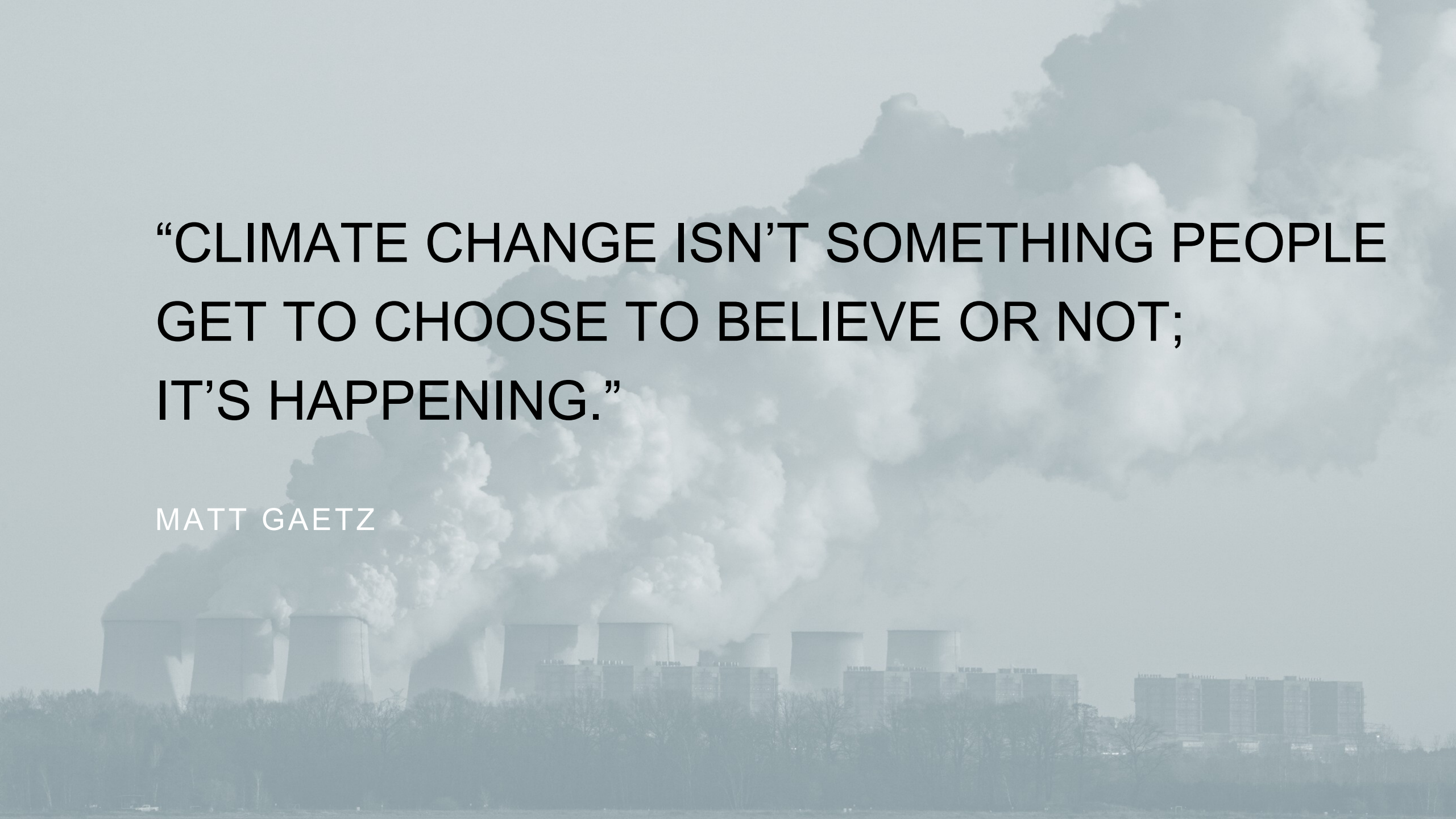


EMPHASIS ON HUMAN CONTRIBUTION

Some people are quick to point out that climate change is a natural phenomenon. However, this is untrue.

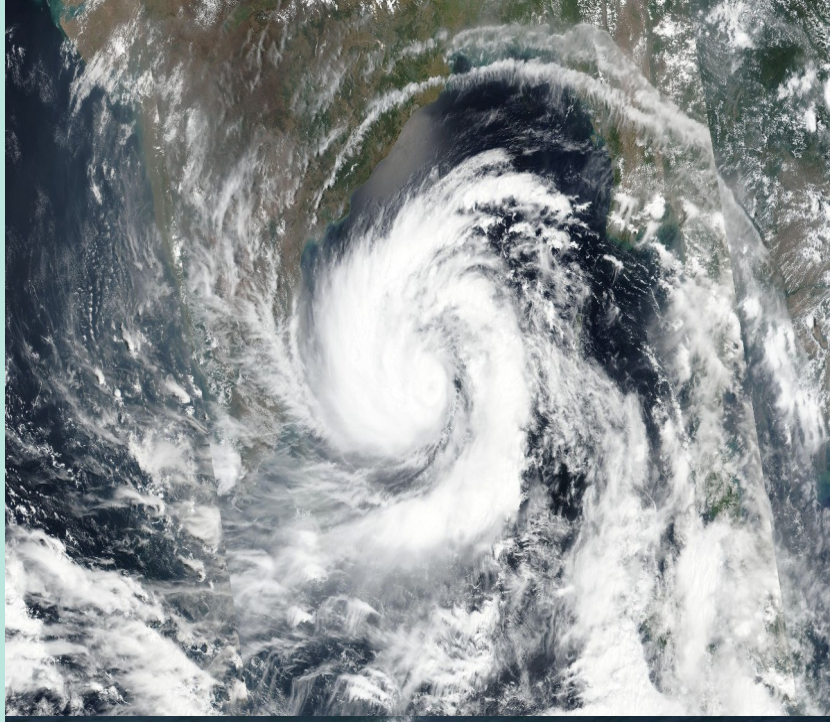
Scientists have considered both causes and have concluded that **without human contribution, modern-era climate change could not have occurred.**

It's time we take responsibility.



“CLIMATE CHANGE ISN’T SOMETHING PEOPLE GET TO CHOOSE TO BELIEVE OR NOT; IT’S HAPPENING.”

MATT GAETZ





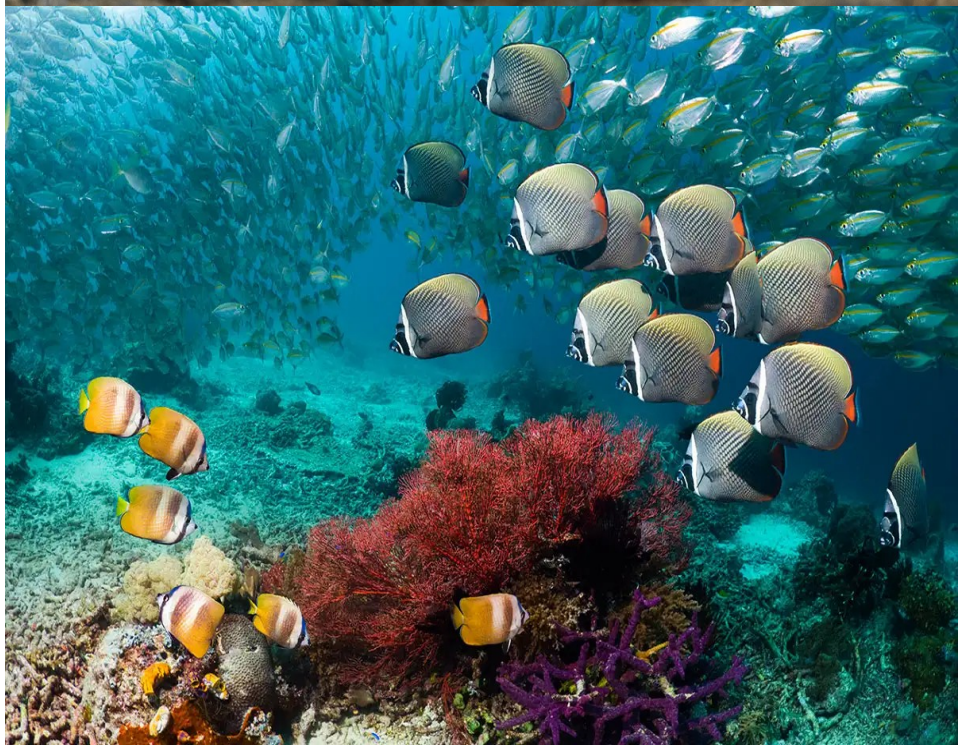
IMPACT ON THE ENVIRONMENT

- **Rise in Temperature**
- **Season Shift:** Seasons come and go earlier or later than their expected time
- **Increased and More Extreme Natural Disasters:**
  - Heavy rain causing floods
  - Severe droughts hampering agricultural productions
  - Powerful cyclones and tsunamis destroying lives and property

IMPACT ON ITS ELEMENTS

- **Water:** Changes in water temperature, water levels, and chemical compositions
- **Soil:** Soil is becoming less fertile and polluted with dangerous chemicals
- **Living Organisms:** While the effects on living organisms are many, the prime concern is that many of them are going extinct
- **Food:** Food is becoming contaminated with hazardous elements, and in some cases, food production is getting lower
- **Human Health:** Various diseases are increasing; ozone layer depletion is increasing the chances of skin cancer





SO, WHAT DO WE DO?

REDUCE, REUSE, RECYCLE

Production takes a lot of energy. By implementing 3Rs, we can reduce our carbon footprint.

SAVE ENERGY AT HOME

Turn things off when not in use and use energy saving equipment.

CHANGE TRANSPORT METHODS

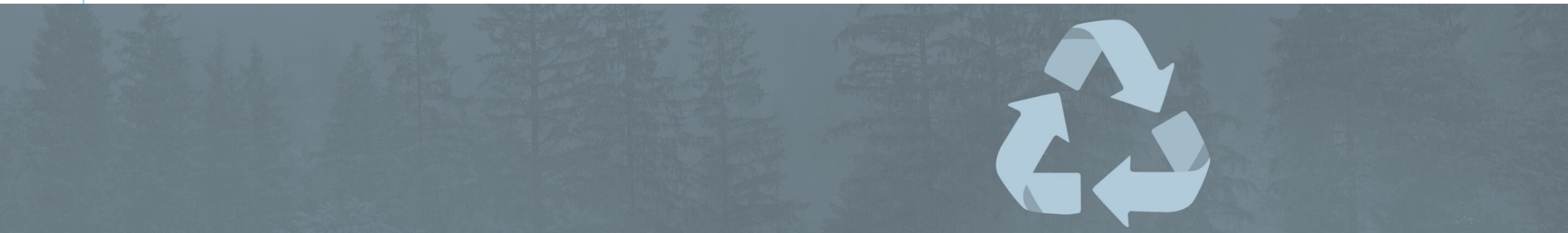
Use public transport more instead of personal cars. Or just use a bicycle.

PLANT MORE TREES

Trees and forests must be saved at all costs. Plant more trees wherever possible.

EDUCATE PEOPLE AROUND US

Awareness is a must. People should be educated about climate change and its dangers.



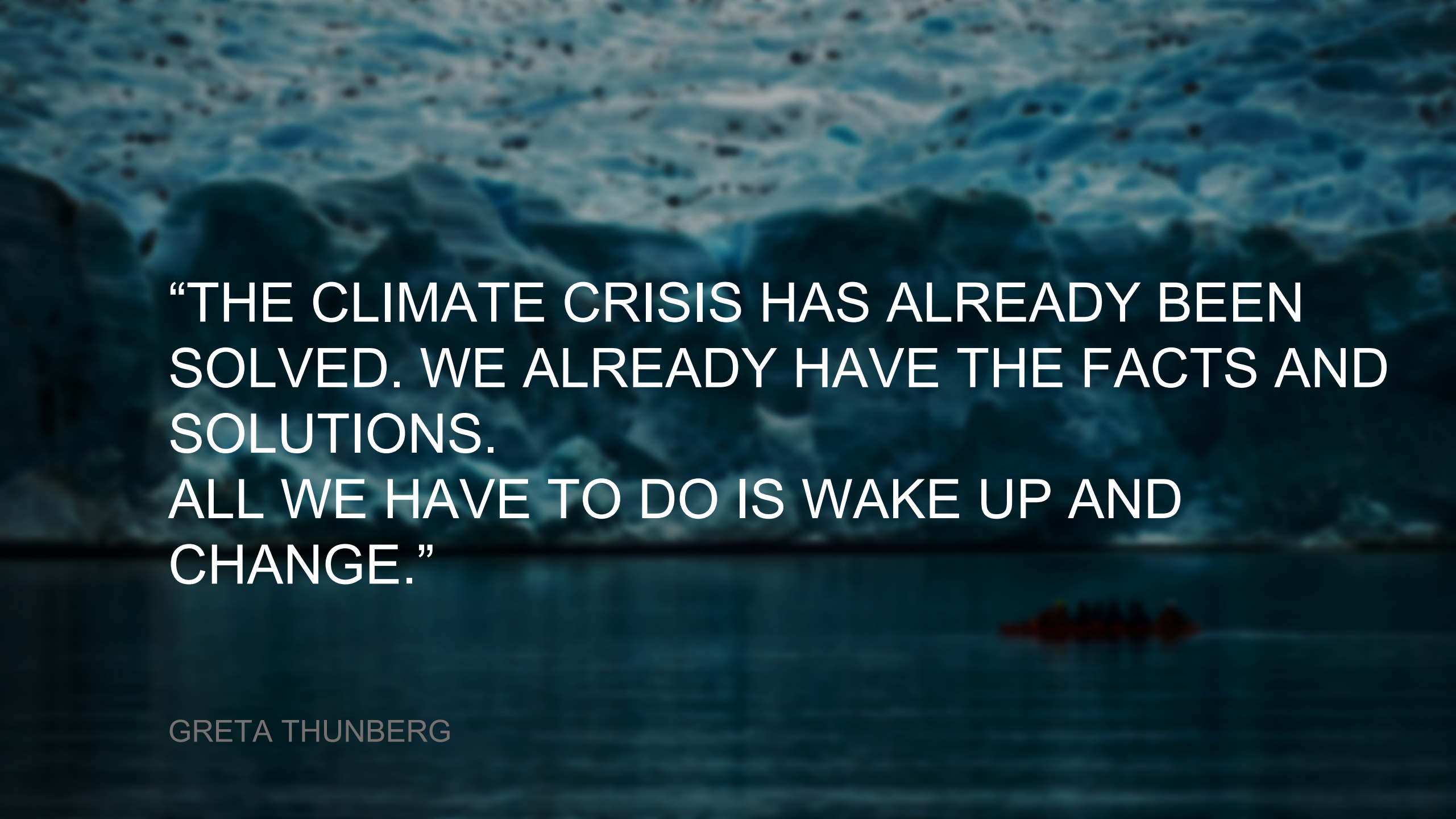
AREAS IN WHICH WE SHOULD FOCUS MOST

REDUCING THE PRODUCTION OF GREENHOUSE GASES

Greenhouse gases are one of the **key reasons for climate change**. So, they are one of the first things that should be addressed.

USING RENEWABLE ENERGY SOURCES

Producing energy using fossil fuels creates a lot of problems. They produce GHGs, and there's also the fact that **we will run out of them soon**. The use of renewable energy everywhere should be made a priority.


“THE CLIMATE CRISIS HAS ALREADY BEEN SOLVED. WE ALREADY HAVE THE FACTS AND SOLUTIONS.
ALL WE HAVE TO DO IS WAKE UP AND CHANGE.”
GRETA THUNBERG

REFERENCES

https://www.climatewatchdata.org/ghgemissions?end_year=2020&start_year=1990

Perkins et al (2013), Economics of Development, seventh Edition

Cunningham Willum, Principles of environmental, second edition

http//gadflyer.com

http://www.nature.com/scitable/knowledge/library/global-atmospheric-change-and-animal-populations

https://www.epa.gov/ghgemissions/sources-greenhouse-gas emissions#:~:text=Carbon%20dioxide%20(CO2)%20makes,natural%20gas%2C%20to%20produce%20electricity